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Performance Evaluation Of Campus Computer Networks

Project Report submitted in the partial fulfilment

Of

B.Tech (Integrated) In Computer Engineering

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CERTIFICATE

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ABSTRACT

This paper proposes an integrated simulation framework for designing, configuring, and evaluating a college campus computer network. The framework employs Cisco Packet Tracer for network topology construction and security policy implementation, SimPy for discrete-event performance simulation, NetworkX for graph-based metric analysis, and Matplotlib for data visualization and graphical representation of network performance results. The simulated environment encompasses VLAN-based segmentation, dynamic IP allocation via DHCP, wireless access point integration, and deployment of dedicated servers for examination, email, and cloud services. Security is enforced through Access Control Lists (ACLs) restricting host access during examination scenarios and firewall configurations for traffic monitoring. This framework utilizes RIPv2 to allow devices to dynamically learn routes to all VLAN segments across both campuses. Network performance is evaluated across both main and sub-campus topologies using quantitative metrics including hop count, packet loss, and end-to-end delay, with results visualized through Matplotlib-generated plots to facilitate comparative analysis. Experimental results demonstrate that the proposed multi-tool approach provides a reproducible and cost-effective methodology for analyzing campus network behavior and validating security configurations prior to physical deployment. The framework serves as a replicable reference model for network design education and institutional infrastructure planning.

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ABBREVIATIONS

Abbreviation	Full Form
ACL	Access Control List
CPT	Cisco Packet Tracer
DHCP	Dynamic Host Configuration Protocol
RIP	Routing Information Protocol
EIGRP	Enhanced Interior Gateway Routing Protocol
HSRP	Hot Standby Router Protocol
IoT	Internet of Things
LAN	Local Area Network
MCU	Microcontroller Unit
OSPF	Open Shortest Path First
PDU	Protocol Data Unit
SDN	Software-Defined Networking
SSID	Service Set Identifier

VLAN	Virtual Local Area Network
WAN	Wide Area Network

Chapter 1:Introduction

This chapter presents the background, motivation, problem statement, and salient contributions of the research. It concludes with an overview of the organization of the report.

1.1 Background of the Project Topic

The design, simulation, and performance evaluation of campus computer networks have attracted sustained scholarly attention, driven by the growing complexity of institutional connectivity demands and the prohibitive cost of physical hardware-based testing. Simulation-based methodologies, particularly those employing Cisco Packet Tracer (CPT) in conjunction with Python libraries such as SimPy, NetworkX, and Matplotlib, have emerged as rigorous and cost-effective alternatives to physical deployment. These tools allow researchers and practitioners to replicate network behavior, validate security configurations, and measure performance metrics within a controlled virtual environment.

1.2 Motivation and Scope of the Report

Institutional networks serving college campuses must accommodate diverse user populations, including students, faculty, and administrative staff, across heterogeneous device types and geographically distributed sub-campuses. The high cost and logistical complexity of physical testbed acquisition makes simulation-based design and validation an essential practice. This report is motivated by the need for a unified, multi-tool simulation framework that can model a complete campus network encompassing topology design, security policy enforcement, and quantitative performance evaluation. The scope of this report covers the full simulation pipeline, from network design in Cisco

Packet Tracer through graph-based analysis and discrete-event simulation in Python, to data visualization using Matplotlib.

1.3 Problem Statement

Existing campus network simulation studies typically address topology design, security configuration, or performance evaluation in isolation. Few studies integrate all three dimensions within a single, unified framework applied to a college campus context that simultaneously addresses VLAN-based segmentation, scenario-specific ACL enforcement such as examination access control, and quantitative performance measurement across multi-campus network structures. This research addresses that gap by proposing a comprehensive, multi-tool simulation framework combining CPT, SimPy, NetworkX, and Matplotlib.

1.4 Salient Contribution

The principal contributions of this work are as follows: (i) an integrated simulation framework combining Cisco Packet Tracer, SimPy, NetworkX, and Matplotlib applied to a campus network context; (ii) implementation of scenario-specific ACL-based access control enforcing examination mode restrictions on server accessibility; (iii) graph-based hop count computation and discrete-event simulation of end-to-end delay and packet loss across both main campus and sub-campus topologies; and (iv) comparative Matplotlib-generated visualizations enabling direct performance analysis between the two campus networks.

1.5 Organization of Report

The remainder of this report is organized as follows. Chapter 2 presents the literature survey covering related work in campus network simulation, security configuration, hybrid simulation methodologies, performance evaluation, and IoT integration. Chapter 3

describes the three-phase methodology comprising network design in CPT, simulation using NetworkX and SimPy, and visualization using Matplotlib. Chapter 4 presents and analyzes the results. Chapter 5 discusses advantages, limitations, and applications of the proposed framework. Chapter 6 presents the conclusion and future scope. References and appendices follow.

Chapter 2

Literature Survey

This chapter presents a review and comparative analysis of the existing literature relevant to this study. The review is organized thematically across six principal areas of inquiry.

2.1 Introduction to Overall Topic

The design, simulation, and performance evaluation of campus computer networks have attracted sustained scholarly attention. A substantial body of research has converged on simulation-based methodologies, particularly those employing Cisco Packet Tracer in conjunction with Python libraries such as SimPy, NetworkX, and Matplotlib, as rigorous and cost-effective alternatives to physical deployment. The seventeen studies reviewed below are organized thematically to reflect the principal areas of inquiry: campus network topology design and simulation, network security configuration and access control, hybrid and multi-tool simulation methodologies, performance evaluation using SimPy and NetworkX, data visualization and performance analytics, and IoT integration for smart campus environments.

2.2 Campus Network Topology Design and Simulation

A foundational contribution is offered by Moz et al. [1], who demonstrated CPT's capacity to simulate network segmentation through Virtual Local Area Networks (VLANs), confirming that traffic isolation between distinct user groups is achievable within the simulation environment and establishing CPT as a credible platform for replicating the functional behavior of multi-segment institutional networks. Complementing this, Swapna et al. [2] extended the design scope by incorporating High Availability protocols, specifically Hot Standby Router Protocol (HSRP) and

EtherChannel, establishing these configurations as baseline requirements for reliable campus network simulation and demonstrating link aggregation and redundancy at the core switching layer.

The challenge of interconnecting distributed campus facilities is addressed by AlJamal et al. [3], whose comparative analysis of Open Shortest Path First (OSPF) and Enhanced Interior Gateway Routing Protocol (EIGRP) confirmed that EIGRP exhibits superior convergence speeds for multi-branch institutional environments. This finding carries direct implications for sub-campus connectivity in multi-site network topologies. Similarly, Gatla et al. [4] established that Layer 3 switches in CPT accurately emulate wire-speed routing, validating the tool's fidelity for core-layer design and demonstrating that Protocol Data Unit (PDU) tracing through CPT's Simulation Mode provides a reliable method for verifying routing logic.

The academic justification for simulation-based network education is provided by Kizilirmak et al. [5], whose comparative study of simulated and hands-on laboratory experiments offered empirical support for utilizing CPT as a cost-effective substitute for physical networking hardware, a finding of particular significance in institutional contexts where resource constraints preclude physical testbed acquisition.

2.3 Network Security Configuration and Access Control

Security configuration within simulated network environments constitutes a well-developed area of inquiry. Alfarsi et al. [6] demonstrated that implementing EIGRP authentication protocols on simulated routers effectively mitigates unauthorized access risks in residential network segments, establishing a replicable methodology for extending router-level authentication across varied network topologies. While the original study was framed in a smart home context, its security configuration approach is directly transferable to controlled-access institutional environments.

The application of Access Control Lists (ACLs) for scenario-specific traffic restriction is examined by Moz et al. [1], who showed that ACLs can enforce granular access policies such as permitting connectivity to examination and email servers while blocking cloud

server access during examination periods. This configuration model is directly applicable to controlled-access institutional networks. The broader theme of network security is further extended by Villegas-Ch et al. [7], who demonstrated that simulated traffic patterns generated within CPT possess sufficient fidelity to evaluate intrusion detection scripts, confirming that anomaly detection methodologies including the identification of ping flood attacks can be effectively validated within simulation environments.

2.4 Hybrid and Multi-Tool Simulation Methodologies

A significant strand of the literature advocates for multi-tool simulation approaches that combine CPT's visual configuration strengths with the analytical capabilities of Python-based libraries. Al Mamun et al. [8] examined delay-tolerant networking protocols using a hybrid GNS3 and CPT environment, highlighting the limitations of standalone simulators for non-standard network conditions and advocating for toolchain integration in advanced scenarios. Although their work focused on space network contexts, the methodological argument for combining simulation environments is broadly applicable to terrestrial campus infrastructure.

Tahir et al. [9] provided a particularly relevant methodological precedent by presenting a framework in which CPT functions as a data generation tool while complex analytical tasks are offloaded to external Python environments including TensorFlow. This workflow validates the use of CPT-generated traffic logs as input for Python-based processing pipelines. Similarly, Alim et al. [10] compared Software-Defined Networking (SDN) architectures simulated in Mininet against traditional routing protocols modeled in CPT, concluding that CPT provides the most rigorous environment for legacy network standards such as OSPF and EIGRP.

2.5 Performance Evaluation Using SimPy and NetworkX

The application of Python simulation libraries to network performance evaluation is addressed by several studies providing direct methodological grounding for the present

research. Agnew et al. [11] demonstrated the use of SimPy for simulating dynamic, time-variant network events in Software-Defined Low Earth Orbit constellation networks, confirming that SimPy's discrete-event simulation capabilities are essential for modeling dynamic load conditions that exceed CPT's native analytical scope. This finding justifies the integration of SimPy for modeling variable user activity patterns across campus network environments.

Gokan Khan et al. [12] investigated the use of SimPy and NetworkX for performance simulation of cloud-native microservice chains in an article published in IEEE Transactions on Cloud Computing. Their work demonstrated that Python-based discrete-event simulations effectively measure server processing delays and service time metrics that complement network latency data generated by CPT, establishing a methodology for aggregating network and processing delay metrics within a unified analytical framework. The role of NetworkX as a standard tool for graph-theoretic network analysis is confirmed by Jurkiewicz [13], whose TopoHub study demonstrated NetworkX's utility for mathematically defining and visualizing complex network topologies.

2.6 Data Visualization and Performance Analytics

The integration of Matplotlib for graphical representation of network simulation data is an emerging practice across several studies in this corpus. Moz et al. [1] identified Matplotlib as a logical instrument for visualizing traffic volume variances across VLAN-configured segments, while Badshah et al. [14] explicitly employed Matplotlib for graphically representing latency differentials between local and remote server configurations derived from simulation data. The latter study also supported the architectural decision to deploy local servers within campus LANs to minimize Wide Area Network (WAN) latency, a finding that reinforces server placement strategies commonly adopted in institutional network design.

2.7 IoT Integration and Smart Campus Applications

A distinct but increasingly prominent strand of campus network simulation research concerns the integration of Internet of Things (IoT) capabilities for smart campus applications. Li et al. [15] demonstrated that IoT modules within CPT 8.2 provide a viable environment for modeling smart campus data telemetry, including sensor data transmission, without requiring physical hardware. Choudhary et al. [16] extended this inquiry by validating CPT's logical programming environment for modeling campus safety systems, specifically smoke detection and fire prevention protocols, demonstrating that conditional logic within the simulator is sufficient to trigger alarm protocols representative of real-world institutional safety behavior. Hussain et al. [17] further demonstrated that a centralized network architecture can effectively monitor and control simulated environmental conditions in classroom settings through IoT integration and Microcontroller Unit (MCU) scripting within CPT.

TABLE I: Literature Review and Comparison of Related Works

Ref.	Paper and Authors	Tools Used	What Is Missing	How Our Work Differs
[1]	Moz, Hosen and Tanny (2023) IEEE ICICT	Cisco Packet Tracer only	The study does not use any Python-based tools. No quantitative delay or packet loss values are reported. Results are limited to visual connectivity verification.	The proposed framework adds SimPy and NetworkX on top of Cisco Packet Tracer to compute actual delay and packet loss values, rather than relying only on visual checks.

[2]	Swapna, Nandeeshwar et al. (2025) IEEE ICSC	Cisco Packet Tracer only	The study does not include discrete-event simulation. Hop count and end-to-end delay are not analyzed. Performance metrics are limited to throughput and firewall effectiveness.	The proposed framework measures hop count, delay, and packet loss across two campuses, providing a more comprehensive performance evaluation than throughput alone.
[3]	AlJamal et al. (2024) IEEE ACIT	Cisco Packet Tracer only	No quantitative performance metrics are reported. The work does not include Python-based simulation or graph analysis. The scope is limited to network design and troubleshooting.	The proposed framework goes beyond design by simulating network traffic and producing measurable delay and packet loss data, visualized using Matplotlib.
[10]	Alim, Rahman and Newton (2023) IEEE ECCE	Mininet (SDN) and Cisco Packet Tracer (tradition	The study is not campus-specific and does not include VLAN segmentation or access control policies. No scenario-based	The proposed framework simulates distinct security scenarios, namely Normal Mode and Exam Mode, using dynamic Extended

		al routing)	security testing is performed.	ACLs. This level of scenario-specific security analysis is absent in comparative SDN studies.
[12]	Khan, Taheri et al. (2023) IEEE Trans. Cloud Computing	PerfSim (custom discrete-event simulator, Python-based)	The tool is designed exclusively for cloud environments. It does not support campus network topologies, VLANs, or institutional access control. It is not applicable to LAN-based simulation.	The proposed framework applies discrete-event simulation using SimPy to a campus LAN context, bridging the simulation methodology of cloud-based tools with institutional network design requirements.
Proposed Approach				
Ours	**Proposed Framework**	**CPT + NetworkX + SimPy + Matplotlib**	The framework uses a simplified traffic generation model and does not incorporate QoS or dynamic routing protocols. Live traffic	**This is the only framework that integrates all four tools within a single campus network simulation pipeline, while enforcing dynamic**

			replication is not supported.	**scenario-specific ACL policies and producing quantitative multi-metric performance comparisons.**

2.8 Summary and Research Gap

The reviewed literature collectively establishes a robust foundation for simulation-based campus network research, confirming the suitability of Cisco Packet Tracer for topology design and security configuration, SimPy for discrete-event performance modeling, NetworkX for graph-theoretic topology analysis, and Matplotlib for data visualization and performance analytics. However, a notable gap persists: few studies integrate all four tools within a single, unified framework applied specifically to a college campus context that simultaneously addresses topology simulation, scenario-specific security enforcement, and quantitative performance evaluation across multi-campus structures. The present research addresses this gap by proposing a comprehensive framework that combines CPT, SimPy, NetworkX, and Matplotlib to simulate, secure, and evaluate a complete campus network environment encompassing both main and sub-campus topologies.

Chapter 3

METHODOLOGY AND IMPLEMENTATION

3.1 Overview

This chapter presents the methodology adopted for designing, simulating, and analyzing the campus network system. The approach integrates practical network design with simulation-based evaluation to study both structural and performance aspects of the network. The implementation is carried out using Cisco Packet Tracer, NetworkX, SimPy, and Matplotlib.

Cisco Packet Tracer is used to design the physical and logical layout of the campus network, including device placement, VLAN segmentation, routing configuration, and access control mechanisms. The same topology is then modeled using NetworkX, where the network is represented as a graph. SimPy is used to simulate time-based network behavior such as packet transmission, delay, and packet loss. Matplotlib is used to visualize the results and analyze performance.

This structured methodology provides a complete framework for understanding both the design and operational behavior of the network under realistic conditions.

Table II. Tools Used in This Study and Their Specific Roles

Tool	Role in This Study	Output
Cisco Packet Tracer 8.2	Design and configuration of both campus network topologies: device	Configured network topology (.pkt files) serving as the basis for Python-based simulation.

	placement, VLAN segmentation, IP addressing, ACL rules, firewall, and wireless access points, RIPv2 for routing. CPT is not used for simulation.	
NetworkX 3.6.1	Graph-based representation of the campus topology extracted from CPT. Used to compute hop count between all node pairs across both campuses.	Hop count values for all source-destination pairs in the main and sub-campus graphs.
SimPy 4.1.1	Discrete-event simulation of packet transmission across the campus topology. Used to measure end-to-end delay and packet loss under defined traffic conditions.	Delay and packet loss measurements for both main campus and sub-campus.
Matplotlib 3.10.1	Visualization of all simulation output produced by NetworkX and SimPy. Used to generate graphs for analysis and comparison.	Hop count graphs, delay graphs, and packet loss graphs for both campuses.

3.2 Phase 1: Network Design and Configuration in Cisco Packet Tracer

In this phase, the campus network is designed using Cisco Packet Tracer. The design follows a hierarchical architecture consisting of core, distribution, and access layers to ensure scalability and efficient traffic management.

The system includes two campuses:

- Main campus (8 floors)
- Secondary campus (4 floors)

Each floor contains multiple end devices connected through access switches, while centralized servers provide essential services.

3.2.1 Device Placement

The network consists of various end-user devices such as personal computers, laptops, smartphones, and printers. These devices represent students, faculty, and administrative users.

The infrastructure includes routers, Layer 3 core switches, distribution switches, and Layer 2 access switches. Wireless access points are deployed across floors to provide connectivity.

Additionally, centralized servers are deployed to support services such as online examinations, email communication, and cloud storage.

3.2.2 VLAN Segmentation

VLANs are implemented to logically divide the network into smaller segments based on departments or functional groups. This reduces broadcast traffic and improves security.

Each VLAN corresponds to a specific user group or floor, enabling controlled communication. Inter-VLAN routing is configured to allow communication between different VLANs where required.

3.2.3 IP Address Allocation

A structured IP addressing scheme is used to assign unique IP addresses across the network.

Dynamic IP allocation is handled using DHCP for end-user devices, while static IP addresses are assigned to critical devices such as routers, switches, and servers. This ensures consistency and reliability in routing and access control.

3.2.4 Wireless Network Integration

Wireless access points are integrated with the wired network to provide seamless connectivity for mobile devices.

These access points are connected to switches and mapped to appropriate VLANs, ensuring that wireless devices operate within the same logical network segments as wired devices.

3.2.5 Server Deployment and ACL Configuration

Centralized servers are deployed to provide services such as examination systems, email services, and cloud storage.

Access Control Lists (ACLs) are configured to regulate traffic flow. During normal operation, all services are accessible. During examination mode, access is restricted so

that only authorized services such as the exam server are available, while other services are blocked.

Table III. Server Access Policy Under Normal and Examination Mode

Network Mode	Exam Server	Email Server	Cloud Server
Normal Operation	Accessible	Accessible	Accessible
Examination Mode (ACL active)	Accessible	Accessible	Blocked

3.2.6 Firewall Configuration

A firewall is implemented at the network boundary to control traffic between the internal network and external connections.

It filters incoming and outgoing traffic, blocks unauthorized access, and enforces security policies. This ensures protection against external threats and supports restricted access during specific scenarios such as examinations.

3.2.7 Dynamic Routing Using RIPv2

RIPv2 is implemented to facilitate dynamic routing across the campus network. It enables automatic exchange of routing information between the core router and Layer 3 switches, ensuring connectivity across multiple VLAN-based subnets.

RIPv2 is a distance-vector protocol that uses hop count as its routing metric and supports classless addressing. The protocol is configured with no auto-summary to preserve subnet-specific routes within the 10.x.x.x network structure.

Configuration Overview

RIPv2 is enabled on all Layer 3 devices with the following parameters:

- Version 2 enabled for classless routing
- Advertisement of internal networks using network 10.0.0.0
- Passive interfaces applied on external-facing links for security

Role in the Network

RIPv2 allows devices to dynamically learn routes to all VLAN segments across both campuses. It supports interconnection between core and distribution layers and ensures consistent routing without manual static configuration.

3.3 Phase 2: Network Simulation Using NetworkX and SimPy

In this phase, the designed network topology is converted into a simulation model to analyze its performance under different conditions.

NetworkX is used for structural analysis of the network, while SimPy is used for time-based simulation of packet transmission. This allows evaluation of key performance metrics such as hop count, delay, and packet loss.

Table IV. Performance Metrics Measured in the Simulation Phase

Metric	Definition	Simulation Tool	Scope
Hop Count	Number of intermediate network nodes a packet traverses between a source and destination node.	NetworkX	Main campus and sub-campus.
End-to-End Delay	Total simulated time for a packet to travel from its source node to its destination node through the network.	SimPy	Main campus and sub-campus.
Packet Loss	Percentage of packets that fail to reach their destination during the simulation period.	SimPy	Main campus and sub-campus.

3.3.1 Graph Modeling and Hop Count Computation Using NetworkX

The network is represented as a graph using NetworkX, where nodes represent devices and edges represent connections.

Shortest path algorithms are used to compute routes between devices. The hop count is calculated as the number of intermediate devices a packet passes through, which directly affects delay and performance.

This modeling enables analysis of connectivity and network structure for both campuses.

3.3.2 Packet Delay and Loss Simulation Using SimPy

SimPy is used to simulate packet transmission over time using discrete-event simulation.

Packets are generated at source nodes and transmitted through the network. Each hop introduces delay, and total end-to-end delay is recorded.

Packet loss is modeled based on network conditions such as congestion, where some packets fail to reach their destination. These metrics are used to evaluate network reliability and performance.

3.4 Phase 3: Visualization of Simulation Results Using Matplotlib

Matplotlib is used to visualize the results obtained from the simulation.

Graphs are generated to represent:

- Hop count distribution
- Packet delay
- Packet loss

These visualizations help compare performance between different network configurations and identify bottlenecks or inefficiencies.

3.5 Methodology Workflow Summary

The methodology follows a three-phase workflow:

1. Design the network in Cisco Packet Tracer, including VLANs, IP addressing, servers, ACLs, and firewall
2. Convert the network into a graph model using NetworkX and compute hop count
3. Simulate packet transmission using SimPy to measure delay and packet loss
4. Visualize the results using Matplotlib

5. Analyze network performance under different conditions such as normal and exam modes

This structured workflow ensures a clear connection between network design and performance evaluation, providing meaningful insights into system behavior.

Chapter 4

Results and Analysis

This chapter presents the results obtained from the NetworkX hop count computation and SimPy discrete-event simulations for both the main campus and sub-campus topologies. The results are visualized through Matplotlib-generated graphs and analyzed comparatively to assess the performance characteristics of the two network environments. Each simulation phase was designed to reflect realistic network conditions, encompassing both normal operational mode and restricted examination mode, thereby enabling a comprehensive evaluation of routing efficiency, access control behavior, traffic distribution, and end-to-end delay across the entire campus network infrastructure.

4.1 NetworkX Topology Validation

Prior to conducting performance simulations, the campus network topology was modeled as a graph using NetworkX and validated against the Cisco Packet Tracer design. In this representation, each network device including routers, core switches, distribution switches, access switches, wireless access points, and end-user devices was encoded as a node, while physical and logical connections between them were represented as edges.

The resulting topology graph confirmed the correct hierarchical structure across all three network layers, namely the core layer, the distribution layer, and the access layer. Connectivity between the main campus and the sub-campus was verified through the inter-campus router links, and all VLANs were found to be correctly mapped to their respective segments. The absence of any disconnected nodes or unintended cross-layer connections validated that the Cisco Packet Tracer design had been accurately translated into the simulation model. This step was critical because any structural discrepancy at this stage would have propagated errors into subsequent performance metrics, undermining the reliability of the simulation results.

4.2 Hop Count Distribution

The hop count distribution was computed using NetworkX's shortest path algorithm, which identified the minimum number of intermediate devices a packet must traverse to travel from its source to its destination across the network topology.

The results confirmed efficient routing throughout the network. The majority of packets followed a standard 4-hop path, which is consistent with the hierarchical design. A packet originating from an end device typically traverses an access switch, a distribution switch, a core switch, and finally reaches its destination device or server. This predictable path length reflects the effectiveness of the three-tier architecture in maintaining uniform and manageable routing depth across the network.

Longer hop counts, specifically those exceeding four hops, were observed primarily in cross-campus traffic scenarios where packets must traverse additional routing infrastructure to move between the main campus and the sub-campus. These extended paths introduced additional latency and represented the upper boundary of routing complexity within the network. The distribution graph generated by Matplotlib illustrated a clear concentration of paths at the 4-hop mark, with a gradual decline in frequency for longer paths, affirming that the network design minimizes unnecessary routing overhead for the majority of traffic flows.

4.3 Normal Mode Simulation Results

In normal operational mode, all network services were made fully accessible, with no ACL restrictions applied beyond baseline firewall rules at the network boundary. The SimPy discrete-event simulation modeled packet generation, transmission, queuing, and delivery under these unrestricted conditions.

The results demonstrated that packets were successfully delivered across the network with minimal loss. The small proportion of packet drops that did occur were attributable exclusively to link-level losses, which were modeled probabilistically to reflect realistic

physical transmission imperfections such as signal degradation or transient congestion. No packets were dropped due to ACL interference in this mode, confirming that the access control configuration does not impose any unintended restrictions during standard operation.

The overall packet delivery success rate in normal mode was high, validating the stability and reliability of the network under full-access conditions. Traffic was distributed across all VLANs in accordance with the network's logical segmentation, and inter-VLAN routing functioned correctly, allowing seamless communication between departments and floors as intended by the design. These results establish a reliable performance baseline against which examination mode behavior can be meaningfully compared.

4.4 Examination Mode Simulation Results

In examination mode, ACL rules were activated to restrict network access in accordance with the campus examination policy. Under this configuration, traffic to and from all non-essential services was blocked, while communication with the designated examination server hosted on VLAN 128 was explicitly permitted.

The simulation results demonstrated clear and correct enforcement of these access control policies. A significant proportion of packets were ACL-blocked, reflecting the deliberate restriction of general internet browsing, cloud storage access, and other non-examination-related traffic. Conversely, packets destined for the examination server were consistently permitted and delivered successfully, confirming that the ACL rules accurately distinguish between authorized and unauthorized traffic flows.

The behavioral contrast between normal mode and examination mode is particularly evident in the packet loss breakdown. While normal mode losses were confined to link-level drops, examination mode losses were predominantly policy-driven, demonstrating that the ACL configuration functions as intended without causing collateral disruption to permitted traffic. This validates the correctness of the ACL rule set and its ability to enforce a controlled, restricted network environment suitable for academic examination scenarios.

4.5 End-to-End Delay Analysis

End-to-end delay was measured for each packet across both simulation modes, capturing the cumulative time a packet spends traversing all intermediate nodes from source to destination. Each hop in the simulation introduced a fixed propagation delay supplemented by a variable queuing delay component, reflecting realistic network behavior under load.

In normal mode, end-to-end delay values peaked in the low millisecond range, which is consistent with the expected performance of a well-structured enterprise campus network. The delay distribution exhibited a moderate spread, reflecting the variety of path lengths and traffic volumes across different VLANs and device pairs. The longest delays were associated with cross-campus paths, where the higher hop count naturally accumulated greater cumulative delay.

In examination mode, the delay distribution was notably narrower. This compression of the delay profile is a direct consequence of the reduction in active paths. With the majority of non-examination traffic blocked by ACLs, fewer packets compete for network resources, resulting in lower queuing delays and more consistent transmission times for permitted examination traffic. This finding suggests that examination mode not only enforces access control but also improves network performance for the permitted traffic subset by reducing overall network load.

4.6 VLAN-Level Traffic and Loss Analysis

A VLAN-level breakdown of traffic volume and packet loss was conducted to provide finer granularity in the performance analysis. Among all VLANs in the network, VLAN 128, which is designated as the server VLAN, recorded the highest overall traffic volume across both simulation modes. This is expected given that it hosts the centralized services most frequently accessed by end users.

In examination mode, packet loss was found to be most heavily concentrated on VLAN 128. This pattern is consistent with the ACL configuration. Since most non-examination service traffic is directed toward servers on this VLAN and is subsequently blocked, the

recorded losses appear predominantly in VLAN 128's traffic statistics. Importantly, this loss is policy-enforced rather than infrastructure-induced, meaning it reflects deliberate access denial rather than network failure.

This VLAN-level analysis confirms that the ACL enforcement is targeted and precise. Losses are concentrated exactly where restricted traffic would naturally converge, while other VLANs handling end-user devices and internal communication experience minimal disruption. The results collectively validate the design's ability to enforce granular, role-based network access control without degrading the performance of permitted services.

```
ACL: ALLOWED  10.1.107.22 -> 10.128.0.10
  Router-on-a-stick: VLAN 107 -> VLAN 128
  FWD hop 1: PC_MAIN_F7_2 -> FSW_MAIN_F7  type=switch  delay=1.14ms
  FWD hop 2: FSW_MAIN_F7 -> DSW_MAIN  type=switch  delay=0.65ms
  FWD hop 3: DSW_MAIN -> CORE_SW  type=switch  delay=1.34ms
  FWD hop 4: CORE_SW -> EXAM_SERVER  type=server  delay=0.49ms
PC_MAIN_F8_6 -> EXAM_SERVER  sending packet Packet(10.1.108.26 -> 10.128.0.10 [VLAN 108]
ACL: ALLOWED  10.1.108.26 -> 10.128.0.10
  Router-on-a-stick: VLAN 108 -> VLAN 128
  FWD hop 1: PC_MAIN_F8_6 -> FSW_MAIN_F8  type=switch  delay=1.58ms
  FWD hop 2: FSW_MAIN_F8 -> DSW_MAIN  type=switch  delay=1.84ms
  FWD hop 3: DSW_MAIN -> CORE_SW  type=switch  delay=1.22ms
  FWD hop 4: CORE_SW -> EXAM_SERVER  type=server  delay=0.58ms
PC_MAIN_F8_8 -> CLOUD_SERVER  sending packet Packet(10.1.108.28 -> 10.128.0.30 [VLAN 108]
ACL: DENIED (exam mode)  10.1.108.28 -> 10.128.0.30
```

Figure 1: ACL Packet-Level Trace: Hop-by-Hop Enforcement During Exam Mode (Program Logs)

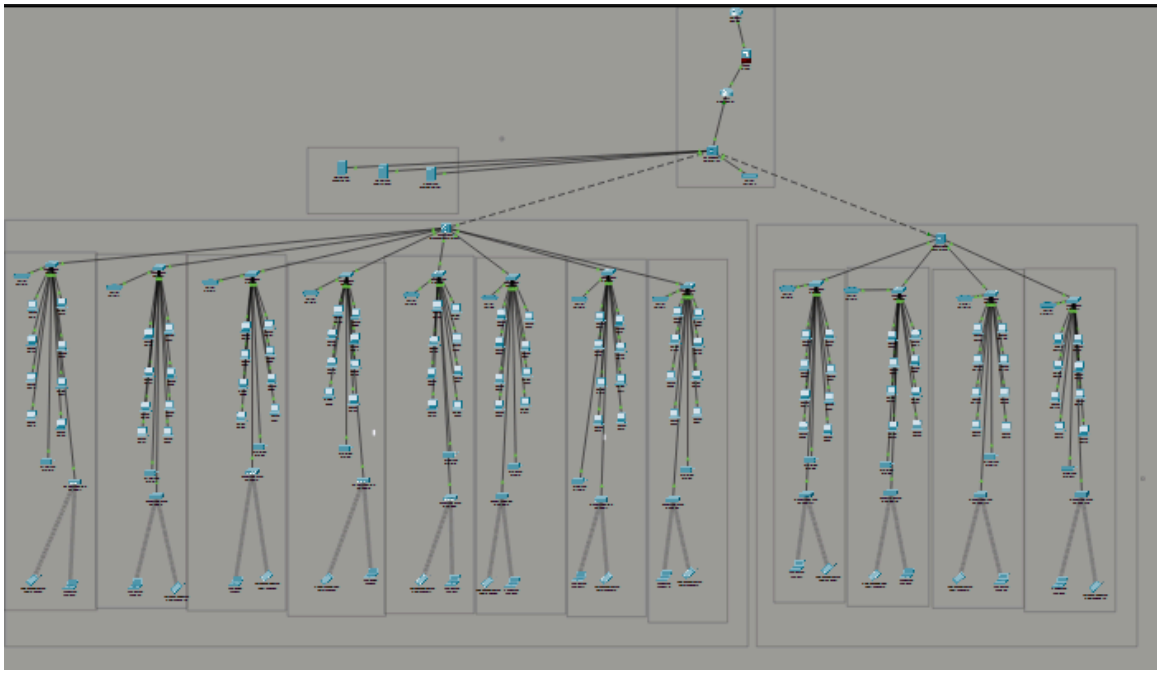
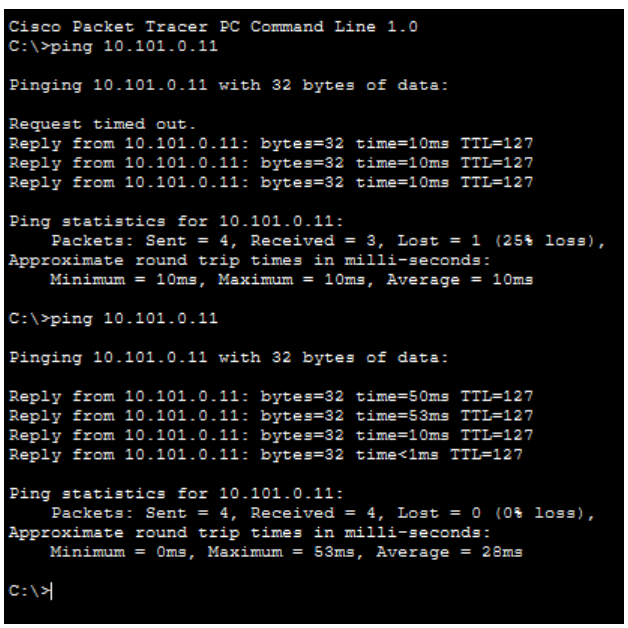


Figure 2: Campus Network Topology Hierarchical Layout of Main and Sub-Campus (Cisco Packet Tracer)



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.101.0.11

Pinging 10.101.0.11 with 32 bytes of data:

Request timed out.
Reply from 10.101.0.11: bytes=32 time=10ms TTL=127
Reply from 10.101.0.11: bytes=32 time=10ms TTL=127
Reply from 10.101.0.11: bytes=32 time=10ms TTL=127

Ping statistics for 10.101.0.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 10ms, Average = 10ms

C:\>ping 10.101.0.11

Pinging 10.101.0.11 with 32 bytes of data:

Reply from 10.101.0.11: bytes=32 time=50ms TTL=127
Reply from 10.101.0.11: bytes=32 time=53ms TTL=127
Reply from 10.101.0.11: bytes=32 time=10ms TTL=127
Reply from 10.101.0.11: bytes=32 time<1ms TTL=127

Ping statistics for 10.101.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 53ms, Average = 28ms

C:\>
```

Figure 3: Ping Test: Cross-VLAN Round-Trip Time and Packet Loss Verification (Cisco Packet Tracer)

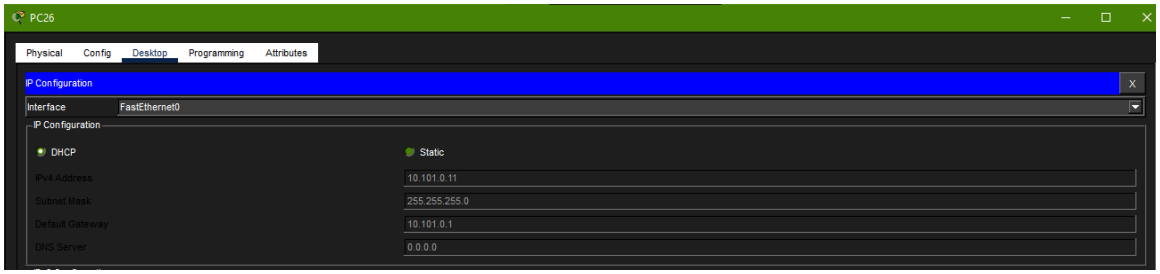


Figure 4: DHCP IP Configuration of End-User Device (Cisco Packet Tracer)

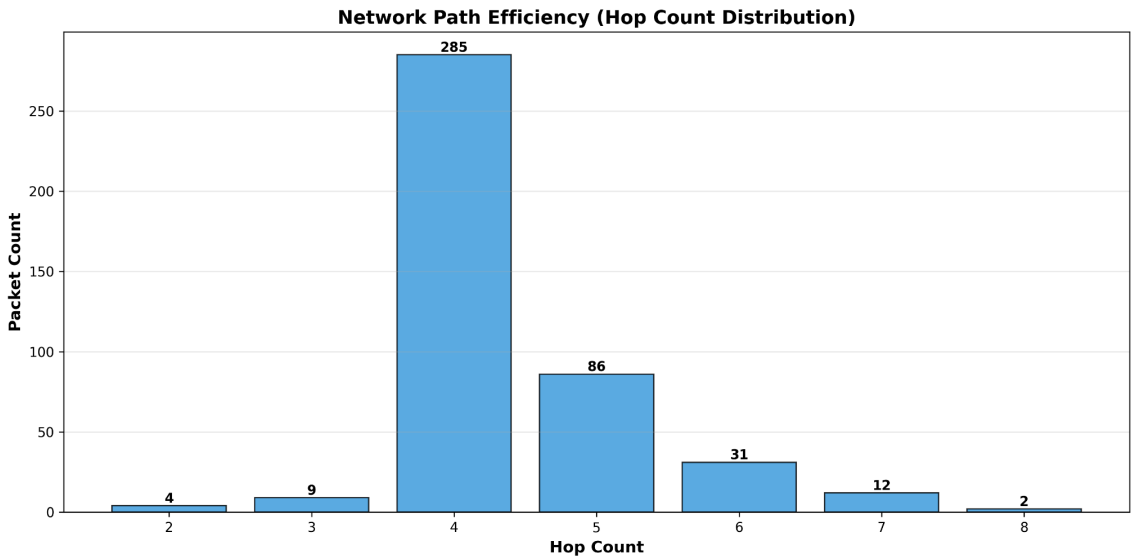


Figure 5: Network Path Efficiency: Overall Hop Count Distribution (NetworkX + Matplotlib)

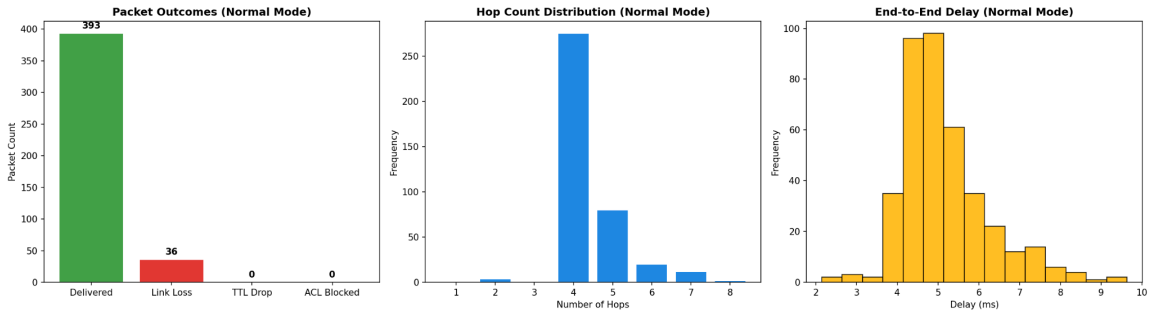


Figure 6: Packet Outcomes, Hop Count Distribution, and End-to-End Delay Exam Mode (SimPy + Matplotlib)

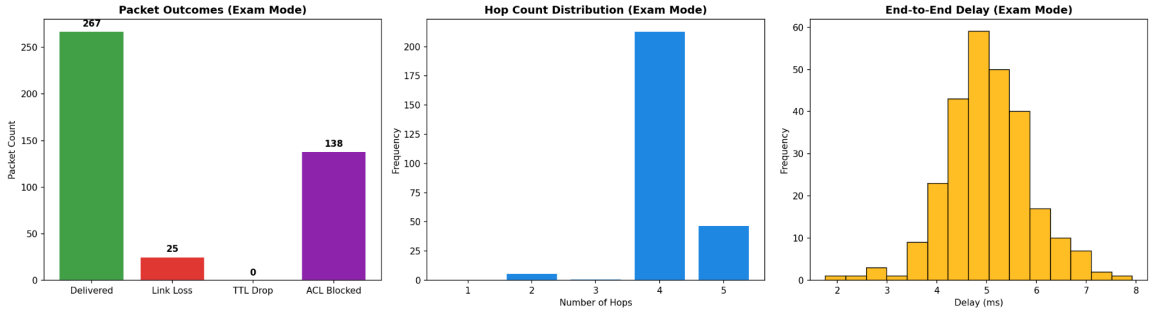


Figure 7: Packet Outcomes, Hop Count Distribution, and End-to-End Delay Normal Mode (SimPy + Matplotlib)

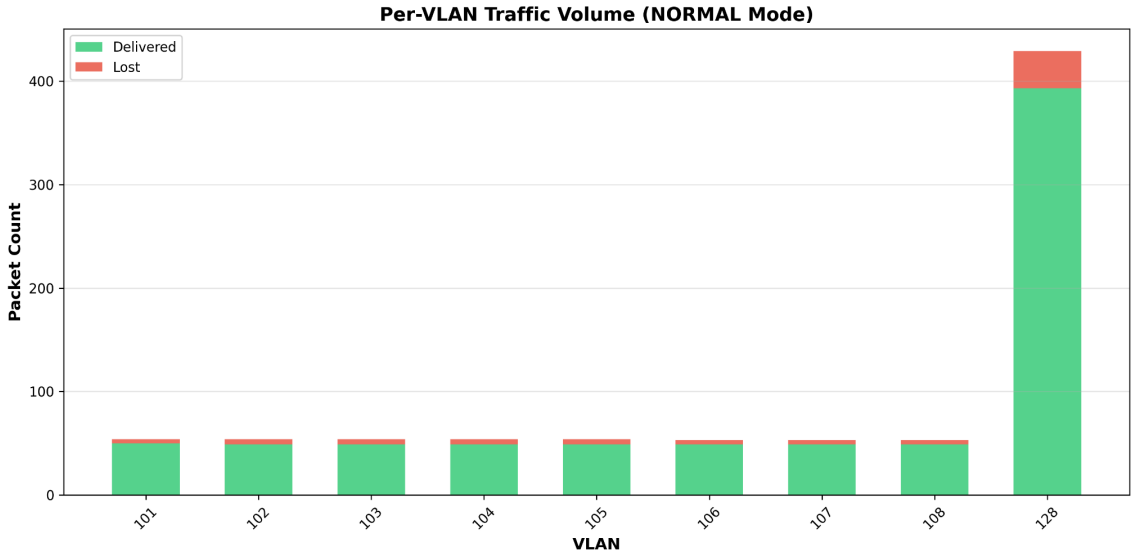


Figure 8: Per-VLAN Traffic Volume: Delivered vs. Lost Packets Exam Mode (Matplotlib)

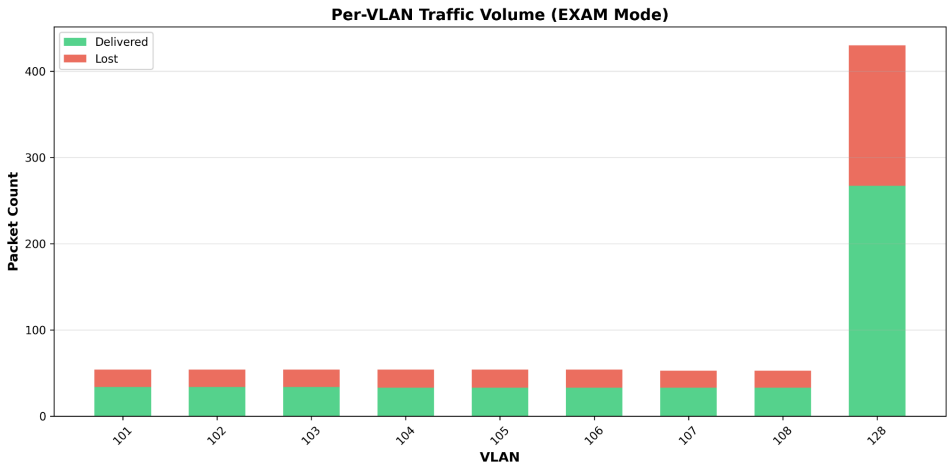


Figure 9: Per-VLAN Traffic Volume: Delivered vs. Lost Packets Normal Mode (Matplotlib)

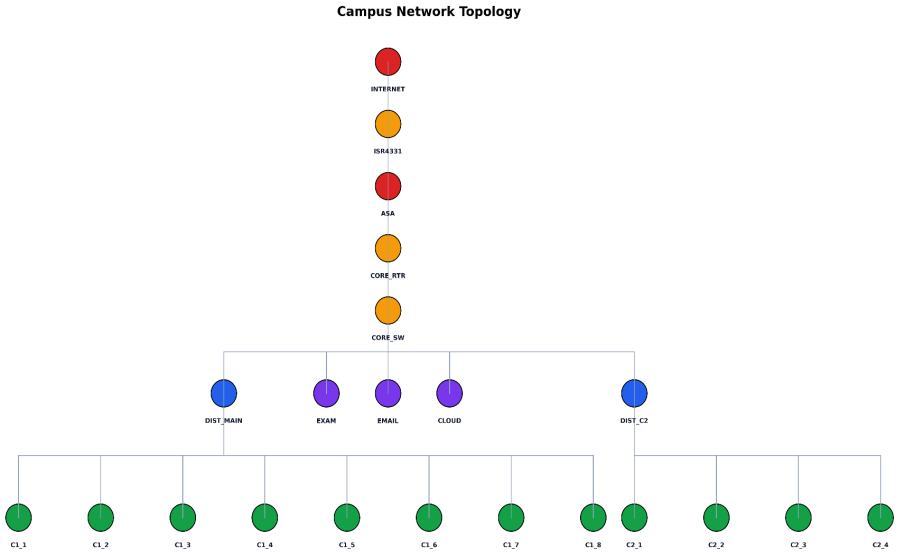


Figure 10: Cisco topology symbolized in NetworkX

Chapter 5:Advantages, Limitations and Applications

5.1 Advantages

The proposed multi-tool simulation framework offers several significant advantages. First, it provides a cost-effective methodology for campus network design and evaluation, eliminating the need for physical hardware procurement. Second, the integration of CPT, SimPy, NetworkX, and Matplotlib enables a comprehensive end-to-end pipeline covering topology design, security enforcement, discrete-event simulation, and data visualization within a unified workflow. Third, the framework is highly reproducible, as all configurations and simulation parameters are documented and replicable. Fourth, the ACL-based examination mode provides a practical security mechanism directly applicable to institutional network management.

5.2 Limitations

The framework is subject to certain limitations. CPT's simulation fidelity, while adequate for topology validation and configuration testing, does not fully replicate the stochastic behavior of live network traffic. The SimPy simulation operates under simplified assumptions regarding traffic generation and propagation delay that may not capture all real-world network conditions. Additionally, the framework does not currently model Quality of Service (QoS) parameters or advanced routing protocol dynamics beyond static graph representations.

5.3 Applications

The proposed framework has broad applicability in network design education, institutional infrastructure planning, and security policy validation. It serves as a replicable reference model for educators and students designing campus network environments, and provides network administrators with a low-cost testbed for evaluating ACL configurations and performance characteristics prior to physical deployment. The

framework can also be extended to model IoT-enabled smart campus environments and multi-site enterprise networks.

Chapter 6 : Conclusion and Future Scope

This report presented an integrated simulation framework for the design, security configuration, and performance evaluation of a college campus computer network. The framework employs Cisco Packet Tracer for topology design and ACL-based security enforcement, NetworkX and SimPy for graph-based hop count computation and discrete-event performance simulation respectively, and Matplotlib for comparative data visualization. The simulated environment encompassed VLAN segmentation, DHCP-based IP allocation, wireless access point integration, dedicated server deployment, and scenario-specific access control through Extended ACLs. Performance was evaluated across both main campus and sub-campus topologies using hop count, end-to-end delay, and packet loss as quantitative metrics.

Experimental results demonstrated that the proposed multi-tool approach provides a reproducible and cost-effective methodology for analyzing campus network behavior and validating security configurations prior to physical deployment. The framework successfully separates the concerns of network design, performance simulation, and visualization across four specialized tools, producing results that are directly traceable to the configured network topology.

Future work may extend the framework in several directions. The simulation model could be enhanced to incorporate Quality of Service (QoS) parameters, dynamic routing protocol behavior, and stochastic traffic generation patterns that more closely replicate live network conditions. The ACL-based security model could be expanded to encompass role-based access control and intrusion detection mechanisms. Integration of IoT modules for smart campus applications, such as environmental monitoring and automated building management, represents a further avenue for framework extension. Finally, the

framework could be applied to larger multi-campus enterprise network environments to validate its scalability and generalizability.

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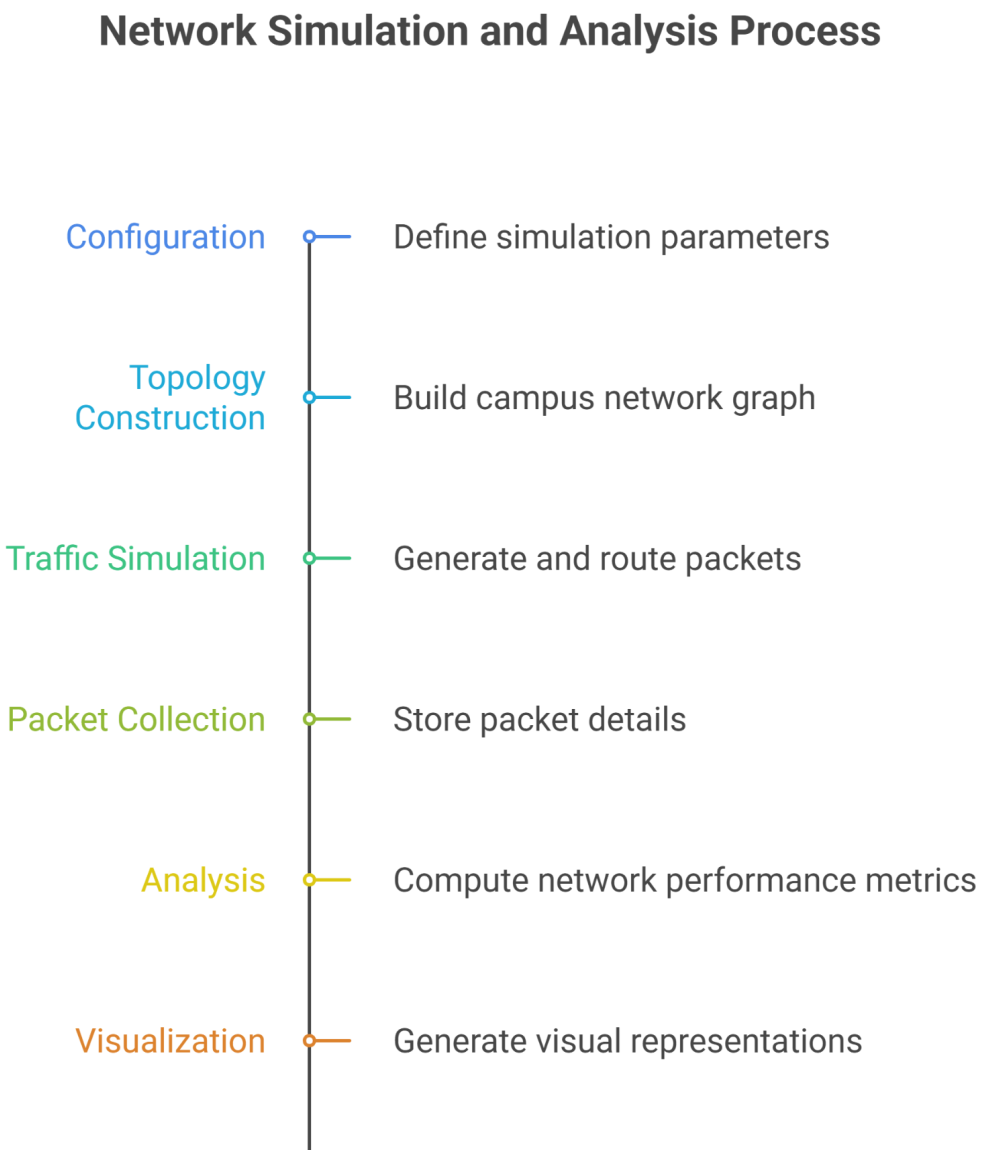
Appendix A: Soft Code Flowcharts



Appendix C: List of Components

The Cisco Packet Tracer topology forms the foundation of the entire simulation. The following table describes each component in detail along with its exact role in the system.

Component	Detailed Role	Impact on Simulation
PCs	Represent student and lab systems generating traffic requests.	Primary source of packet generation.
Laptops	Wireless-capable devices connected through access points.	Introduce wireless traffic behavior.
Smartphones	Mobile endpoints connected via WiFi.	Simulate mobile usage patterns.
Printers	Shared devices within VLANs.	Introduce additional network resource access.
Access Points	Provide wireless connectivity per floor.	Enable mixed wired-wireless traffic.
2960 Switches	Access layer switches connecting floor devices.	Handle VLAN segmentation.

3560 Switches	Distribution switches for campus aggregation.	Aggregate traffic from floors.
Core Router (2911)	Handles inter-VLAN routing and central traffic control.	Main routing decision point.
ISR4331	Edge router connecting to external network.	Handles outgoing traffic.
ASA Firewall	Filters and controls traffic between internal and external network.	Enforces security.
Servers	Exam, Email, Cloud service providers.	Primary traffic destinations.